LOAN MANAGEMENT SYSTEM (LMS)

HIGH LEVEL DESIGN (HLD) & LOW LEVEL DESIGN (LLD)

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1. INTRODUCTION

1.1 Overview

The Loan Management System (LMS) is an enterprise web-based application designed to manage the complete loan lifecycle in a structured, secure, and role-based manner. The system supports customer onboarding, KYC verification, internal credit scoring, loan application processing, approval workflows, sanction letter handling, and loan disbursement.

The LMS is designed to closely resemble real-world banking and NBFC loan processing systems, ensuring compliance, auditability, and controlled access for all stakeholders.

1.2 Purpose of the Document

This document describes the High-Level Design (HLD) and Low -Level Design (LLD) of the Loan Management System.

The purpose of this document is to:

- Explain system architecture and workflow
- Define roles and responsibilities
- Describe database structure and business logic
- Provide a clear understanding of system behaviour

1.3 Scope of the System

The scope of the LMS includes:

- User registration and authentication
- KYC submission and verification
- Internal scoring and eligibility calculation
- Loan application and document verification
- Multi-level loan approval
- Sanction letter generation and verification
- Loan disbursement and monitoring

1.4 Definitions and Abbreviations

Term	Description
LMS-	Loan Management System
KYC-	Know Your Customer
EMI-	Equated Monthly Instalment
RBAC-	Role-Based Access Control

2. SYSTEM OVERVIEW

2.1 User Roles

The system supports four distinct roles:

1. **Customer** – Applies for loans and tracks status
2. **Verification Team** – Verifies KYC and documents, assigns scores
3. **Manager** – Reviews applications and approves loans under limits
4. **Admin** – Final authority for sanction, disbursement, and system control

Each role has restricted access based on business responsibility.

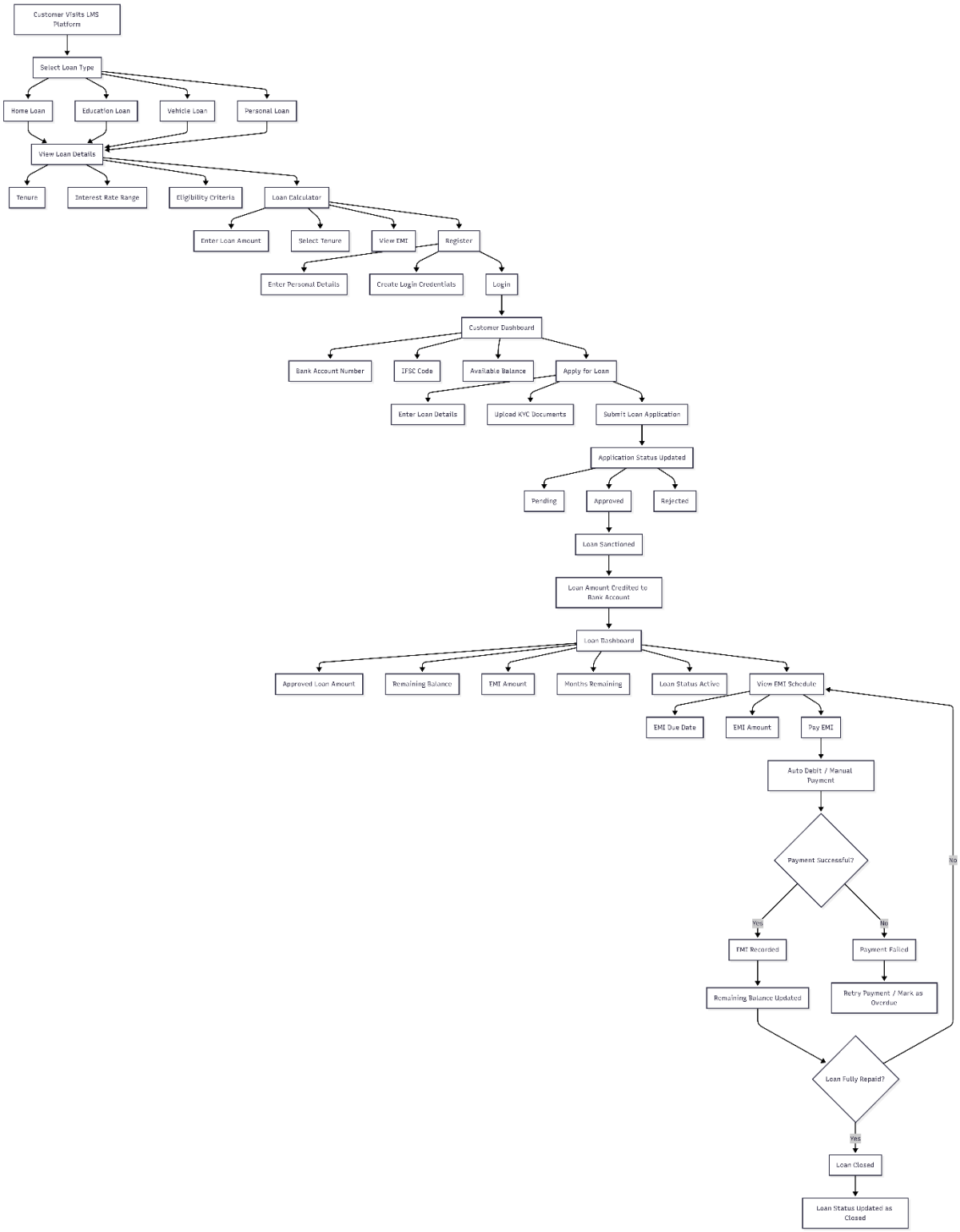
2.2 Business Objectives

The LMS aims to:

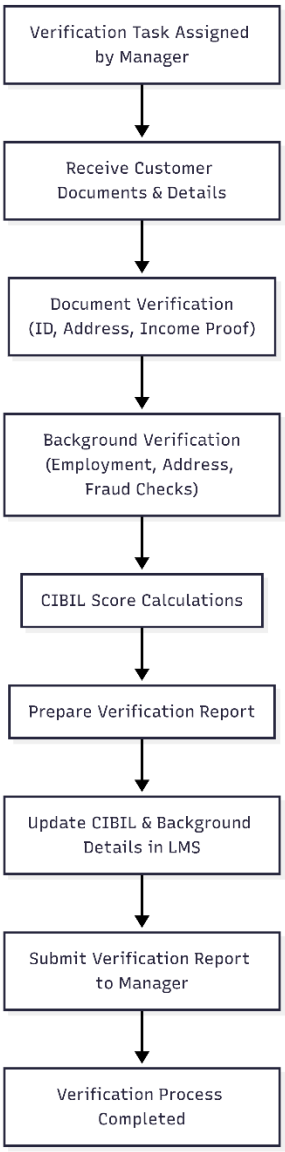
- Automate loan processing
- Reduce manual intervention
- Enforce approval hierarchy
- Maintain audit trails
- Improve transparency for customers

3. System Architecture and roles

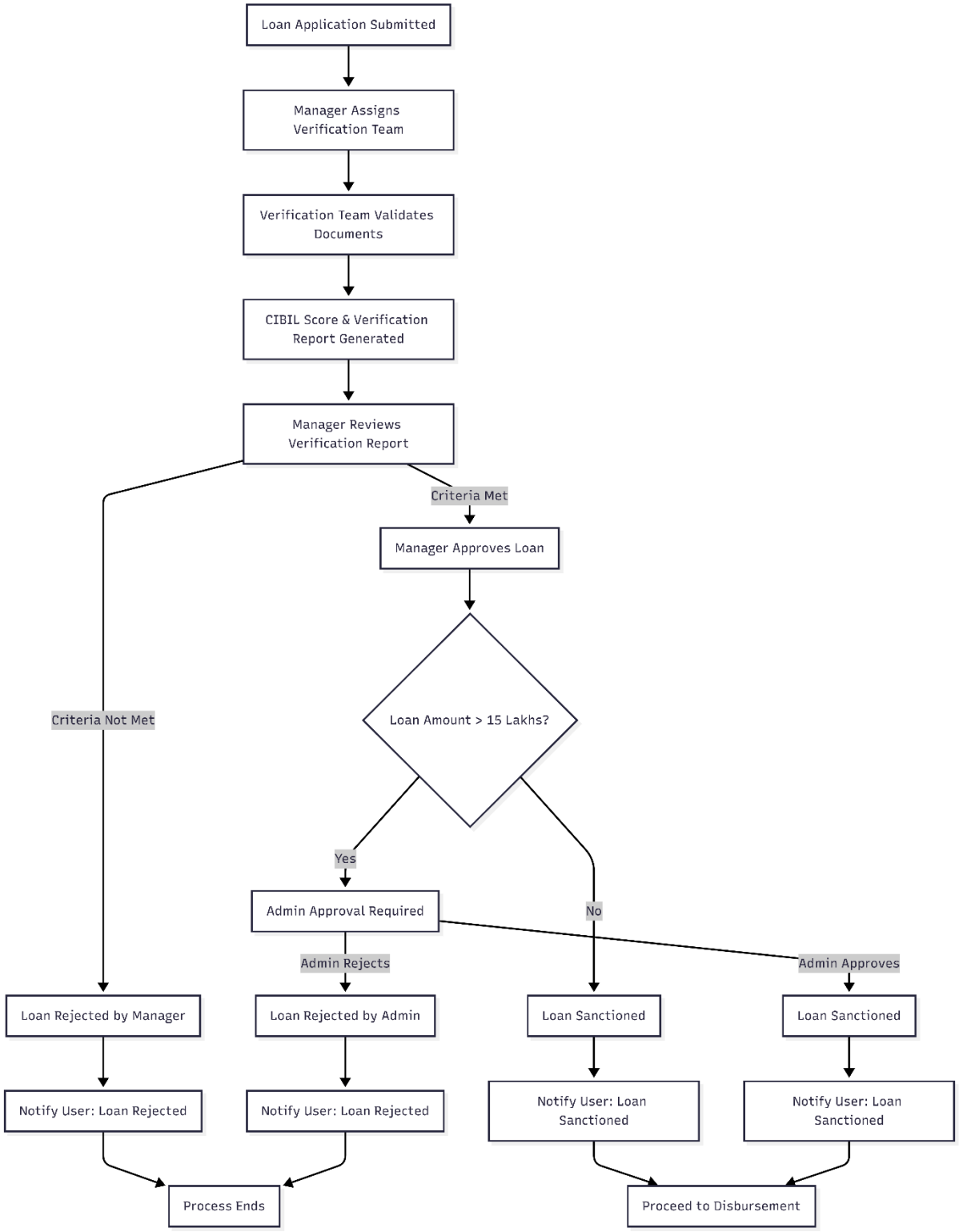
3.1 Customer Flow



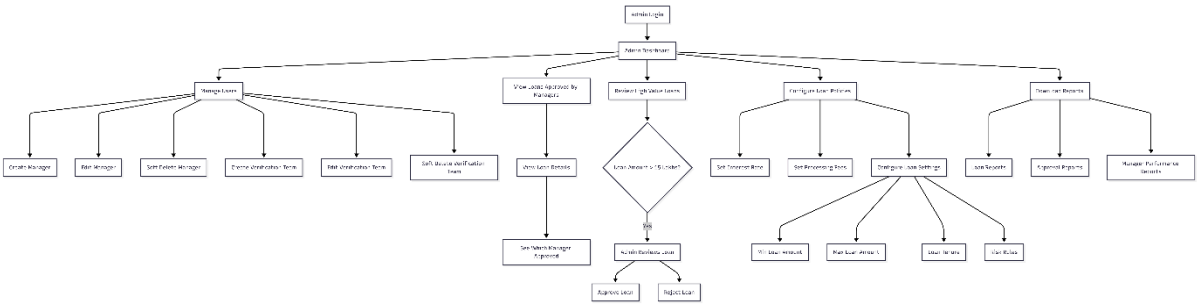
3.2 Verification Team Flow



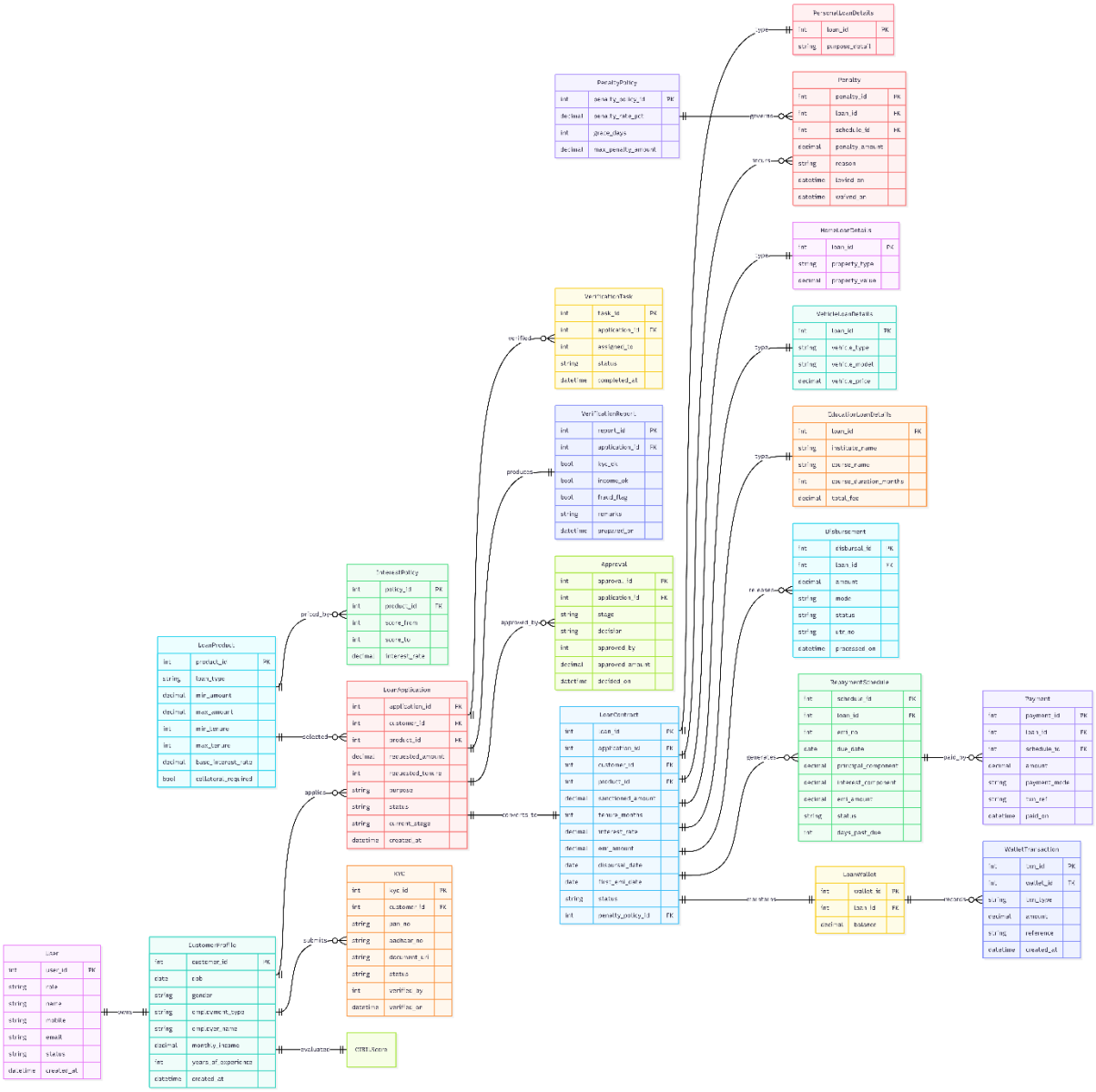
3.3 Manager Flow



3.4 Admin Flow



4.Database Design



5.Backend API and integration

AUTHENTICATION ENDPOINTS

POST /api/auth/register

Description: Registers a new customer with personal details (name, email, password, mobile, DOB, gender, PAN). Creates user account and auto-generates virtual bank account with zero balance.

POST /api/auth/login

Description: Authenticates user (all roles) with email and password. Returns JWT token for secure access to protected endpoints.

CUSTOMER ENDPOINTS

GET /api/customer/dashboard

Description: Returns customer dashboard summary including KYC status, virtual account details, active loans, and next action required.

POST /api/customer/kyc/submit

Description: Submits complete KYC information in one call including personal details (Step 1), employment & income details (Step 2), and address details (Step 3). Sets KYC status to PENDING_VERIFICATION.

GET /api/customer/kyc/status

Description: Returns current KYC verification status (Pending/Approved/Rejected), verification score, and whether customer can apply for loans.

POST /api/customer/loans/apply

Description: Creates a new loan application with loan type, amount, tenure, and purpose. Returns loan ID and prompts for document upload.

POST /api/customer/loans/{loan_id}/documents

Description: Uploads exactly 3 required documents for the specific loan type (e.g., Salary Slip, Bank Statement, Identity Proof for Personal Loan). Accepts multipart form data.

GET /api/customer/loans

Description: Returns list of all loan applications submitted by the customer with their current status.

GET /api/customer/loans/{loan_id}

Description: Returns detailed information about a specific loan including amount, tenure, interest rate, EMI, current status, and complete timeline of approval stages.

GET /api/customer/loans/{loan_id}/sanction-letter

Description: Returns the generated sanction letter PDF URL for download after loan approval.

POST /api/customer/loans/{loan_id}/sanction-letter/upload

Description: Uploads the e-signed sanction letter. Accepts signed PDF file and forwards to manager for signature verification.

GET /api/customer/loans/{loan_id}/emi-schedule

Description: Returns complete EMI repayment schedule showing installment number, due dates, EMI amounts, principal/interest breakdown, payment status, and any penalties.

POST /api/customer/loans/{loan_id}/pay-emi

Description: Processes EMI payment for a specific installment. Deducts amount from virtual account, updates EMI status to PAID, and records transaction.

GET /api/customer/account/transactions

Description: Returns virtual bank account transaction history including loan disbursements, EMI payments, and current balance.

VERIFICATION TEAM ENDPOINTS

GET /api/verification/kyc/pending

Description: Returns list of all KYC submissions pending verification with customer name, PAN, and submission date.

GET /api/verification/kyc/{kyc_id}

Returns complete KYC details for a specific customer including personal, employment, income, and address information for verification.

POST /api/verification/kyc/{kyc_id}/verify

Description: Verifies KYC by assigning scores (0-25 each) for Income, EMI burden, Employment Type, and Experience. Calculates total score and approves/rejects KYC.

GET /api/verification/loans/pending-documents

Description: Returns list of loan applications with uploaded documents pending verification.

GET /api/verification/loans/{loan_id}/documents

Description: Returns all 3 uploaded documents for a specific loan with download URLs for verification team to review.

POST /api/verification/loans/{loan_id}/documents/verify

Description: Verifies each of the 3 uploaded documents individually (Approve/Reject each). Makes overall decision and forwards loan to Manager if all documents approved.

MANAGER ENDPOINTS

GET /api/manager/dashboard

Description: Returns manager dashboard statistics including pending approvals, today's approved/rejected loans, and pending signature verifications.

GET /api/manager/loans/pending-approval

Description: Returns list of loans pending manager approval. Automatically filters to show only loans ≤15 Lakhs (manager approval limit).

GET /api/manager/loans/{loan_id}/review

Description: Returns complete loan details for review including customer info, KYC score, verification scores breakdown, loan amount, tenure, and document verification status.

POST /api/manager/loans/{loan_id}/approve

Description: Approves loan if amount ≤15 Lakhs. Sets approved amount, interest rate, and forwards to Admin for sanction letter generation.

POST /api/manager/loans/{loan_id}/forward-to-admin

Description: Forwards loan to Admin if amount >15 Lakhs (exceeds manager approval authority). Includes manager recommendation and remarks.

POST /api/manager/loans/{loan_id}/reject

Description: Rejects loan application with reason. Ends the loan process and notifies customer.

GET /api/manager/sanction-letters/pending-verification

Description: Returns list of signed sanction letters uploaded by customers pending manager signature verification.

POST /api/manager/sanction-letters/{loan_id}/verify-signature

Description: Verifies customer signature on signed sanction letter. Compares original and signed documents, approves for disbursement if verified.

GET /api/manager/emi/overdue

Description: Returns list of all loans with overdue EMIs for monitoring. Read-only access for manager to track customer repayment issues.

ADMIN ENDPOINTS

GET /api/admin/dashboard

Description: Returns admin dashboard statistics including high-value loans pending approval (>15L), total disbursements today, total active loans, and system health metrics.

GET /api/admin/loans/high-value-pending

Description: Returns list of loans >15 Lakhs pending admin approval that were forwarded by managers.

GET /api/admin/loans/{loan_id}/review

Description: Returns complete loan details for admin review including all verification scores, manager recommendation, and complete customer profile.

POST /api/admin/loans/{loan_id}/approve

Description: Approves high-value loans (>15L). Sets final approved amount and interest rate, triggers sanction letter generation.

POST /api/admin/loans/{loan_id}/reject

Description: Rejects high-value loan with reason. Final rejection authority - ends loan process.

POST /api/admin/loans/{loan_id}/sanction-letter/generate

Description: Generates sanction letter PDF for approved loans. Only Admin has authority to generate official sanction letters. Returns PDF URL for customer download.

GET /api/admin/loans/ready-for-disbursement

Description: Returns list of loans with verified signatures ready for disbursement (sanction letter signed and verified by manager).

POST /api/admin/loans/{loan_id}/disburse

Description: Disburses approved loan amount to customer's virtual bank account. Credits amount, creates transaction record, activates loan, and triggers EMI schedule generation.

GET /api/admin/users

Description: Returns list of all system users (Customers, Verification Team, Managers) with their roles and account status.

POST /api/admin/users/create

Description: Creates new system users (Verification Team members or Managers). Assigns roles and sets initial credentials.

PUT /api/admin/users/{user_id}/status

Description: Enables or disables user accounts. Used to manage system access for staff members.

GET /api/admin/audit-logs

Description: Returns system audit logs for all critical actions (login, KYC submissions, loan approvals, disbursements, EMI payments) with filters for date range, user, and action type.

GET /api/admin/emi/monitoring

Description: Returns comprehensive EMI monitoring dashboard showing all active loans, overdue EMIs, penalty amounts, and escalation cases (3+ consecutive missed EMIs).

POST /api/admin/emi/{emi_id}/apply-penalty

Description: Manually applies penalty to overdue EMI. Records penalty amount and reason in system.

6.Core Modules of Loan Management System (LMS)

The Loan Management System (LMS) is designed as a modular, role-based enterprise application. Each module is responsible for a specific phase of the loan lifecycle and interacts with other modules through controlled workflows and APIs. This modular design improves scalability, maintainability, security, and auditability.

6.1 User Registration & Authentication Module

Description

This module manages customer onboarding, identity creation, and secure authentication. It ensures that each user is uniquely identified and prevents duplicate accounts.

Key Responsibilities

- Register new users with mandatory details
- Enforce **one PAN per user** rule
- Authenticate users using secure credentials
- Generate and manage login sessions
- Support role-based login (Customer, Verification Team, Manager, Admin)

Functional Flow

- User submits registration details
- System validates PAN uniqueness
- User account is created
- Virtual bank account is auto-generated
- User logs in using credentials

Security Controls

- Password hashing (bcrypt/passlib)
- JWT-based authentication
- Account lock after repeated failed attempts
- Role-based access control (RBAC)

Outcome

Only authenticated and authorized users can access the LMS based on their role.

6.2 KYC Management Module

Description

The KYC (Know Your Customer) module ensures regulatory compliance by collecting, validating, and approving customer identity and financial details before loan eligibility.

Key Responsibilities

- Collect customer identity details
- Capture employment and income information
- Record existing EMI obligations
- Maintain KYC status (Pending / Approved / Rejected)

KYC Structure

- **Step 1:** Applicant Details
- **Step 2:** Employment & Income Details
- **Step 3:** Address Details

Verification Flow

- Customer submits KYC
- Verification Team reviews details
- KYC is approved or rejected
- Loan access is enabled only after approval

Outcome

Ensures only verified users can proceed to loan application.

6.3 Risk Assessment & Internal CIBIL Engine

Description

This module evaluates customer creditworthiness using internal scoring logic, similar to a CIBIL score, based on financial stability and repayment capacity.

Scoring Parameters

Each parameter is scored from 0–25:

- Monthly Income
- Existing EMI burden
- Employment Type
- Years of Experience

Responsibilities

- Aggregate verification team scores
- Calculate total risk score (out of 100)
- Determine loan eligibility range (min/max amount)
- Influence interest rate and approval flow

Ownership

- Verification Team provides raw scores
- Manager validates and finalizes eligibility

Outcome

Provides a data-driven foundation for loan approval decisions.

6.4 Loan Application Module

Description

This module allows eligible customers to apply for loans based on approved KYC and risk score.

Supported Loan Types

- Personal Loan
- Home Loan
- Vehicle Loan
- Education Loan

Key Responsibilities

- Capture loan amount, tenure, and purpose
- Auto-calculate EMI and interest rate
- Enforce eligibility limits
- Prevent loan application before KYC approval

Outcome

Creates a formal loan request that enters the approval workflow.

6.5 Document Management Module

Description

This module manages secure upload, storage, and verification of loan-specific documents.

Mandatory Documents (Exactly 3 per Loan Type)

Personal Loan

- Salary Slip
- Bank Statement
- Identity Proof

Home Loan

- Property Documents
- Sale Agreement
- Identity Proof

Vehicle Loan

- Vehicle Quotation
- Dealer Invoice
- Identity Proof

Education Loan

- Admission Letter
- Fee Structure
- Identity Proof

Responsibilities

- Validate file type and size
- Store documents securely
- Provide read-only access to reviewers

Outcome

Ensures authenticity of loan applications before approval.

6.6 Loan Approval & Workflow Module

Description

This module controls the multi-level approval process based on loan amount and risk category.

Approval Logic

- Loans ≤ 15 Lakhs → Manager approval
- Loans > 15 Lakhs → Admin approval

Responsibilities

- Route applications automatically

- Enforce approval hierarchy
- Prevent unauthorized approvals

Outcome

Maintains governance and controlled escalation.

6.7 Sanction Letter Management Module

Description

This module handles legal confirmation of loan approval.

Responsibilities

- Generate sanction letter (Admin only)
- Allow customer to download letter
- Accept digitally signed sanction letter
- Allow Manager to verify signed document

Outcome

Ensures legal acceptance before disbursement.

6.8 Loan Disbursement Module

Description

This module handles the release of loan funds.

Responsibilities

- Disburse approved loan amount (Admin only)
- Credit funds to customer virtual bank account
- Activate loan status

Outcome

Marks the official start of the loan lifecycle.

6.9 EMI Management & Repayment Module

Description

This module manages loan repayment and tracking.

Responsibilities

- Generate EMI schedule automatically

- Track monthly payments
- Apply penalties for overdue EMIs
- Maintain complete EMI history

EMI Status

- Paid
- Pending
- Overdue

Escalation Rule

- 3 consecutive missed EMIs trigger escalation

Outcome

Ensures repayment discipline and financial tracking.

6.10 EMI History & Monitoring Module

Description

This module provides transparency and monitoring of repayment behavior.

Visibility

- Customer: Own EMI history
- Manager: Read-only monitoring
- Admin: Audit and recovery

Stored Information

- Installment number
- Due date
- Payment date
- Paid amount
- Penalty
- Status

Outcome

Supports audit, recovery, and compliance.

6.11 Audit & Logging Module

Description

This module ensures traceability of all critical actions.

Logged Events

- Login attempts
- KYC submissions
- Approvals and rejections
- Sanction generation
- Disbursement
- EMI payments

Responsibilities

- Maintain immutable logs
- Support compliance and audits

Outcome

Ensures accountability and regulatory readiness.

6.12 Administration & Governance Module

Description

This module allows administrative control without interfering in daily operations.

Responsibilities

- Create / disable users
- Modify system rules
- Monitor system activity
- Handle escalations

Outcome

Provides centralized oversight and governance.

7. Frontend UI Interfaces:

https://www.figma.com/make/GuNE6nRzRCRXaQJszuuaqr/Low-Fidelity-Wireframe-for-LMS?p=f&t=yqgOpZuWBQwBFdkU-0

7.1.Customer Registration

FLOW 1: REGISTRATION

CUSTOMER REGISTRATION FORM

Full Name
[Full Name Input]

Mobile Number
[Mobile Number Input]

Date of Birth
[DOB Input]

Gender
[Select Gender]

Email
[Email Input]

PAN Number
[PAN Number Input]

Password
[Password Input]

Confirm Password
[Confirm Password Input]

REGISTER

SYSTEM NOTE:
Virtual Bank Account auto-created after registration

7.2.Customer Login

CUSTOMER LOGIN

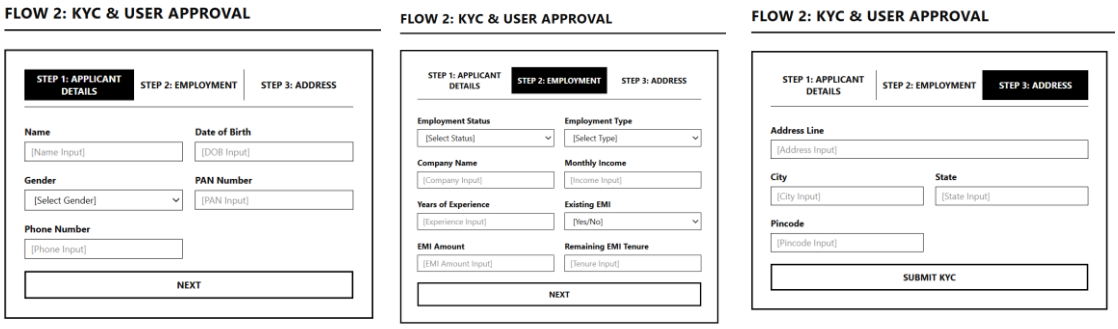
Email
[Email Input]

Password
[Password Input]

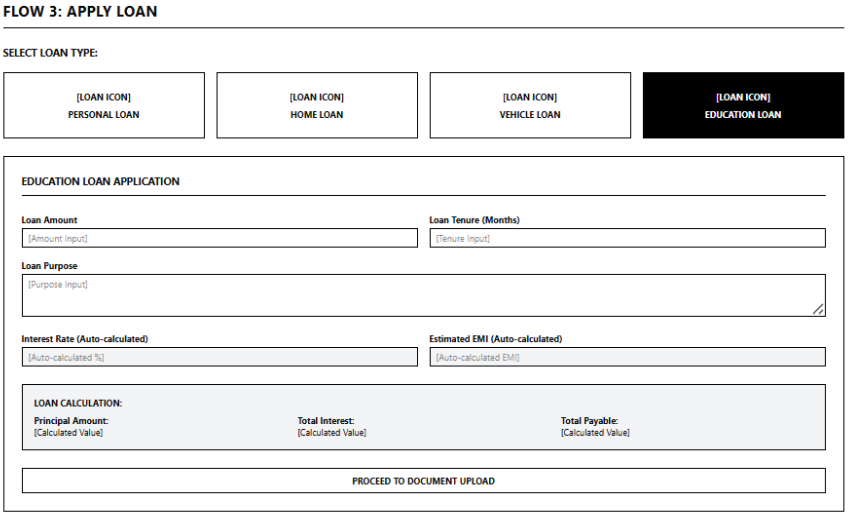
LOGIN

New user? Register here

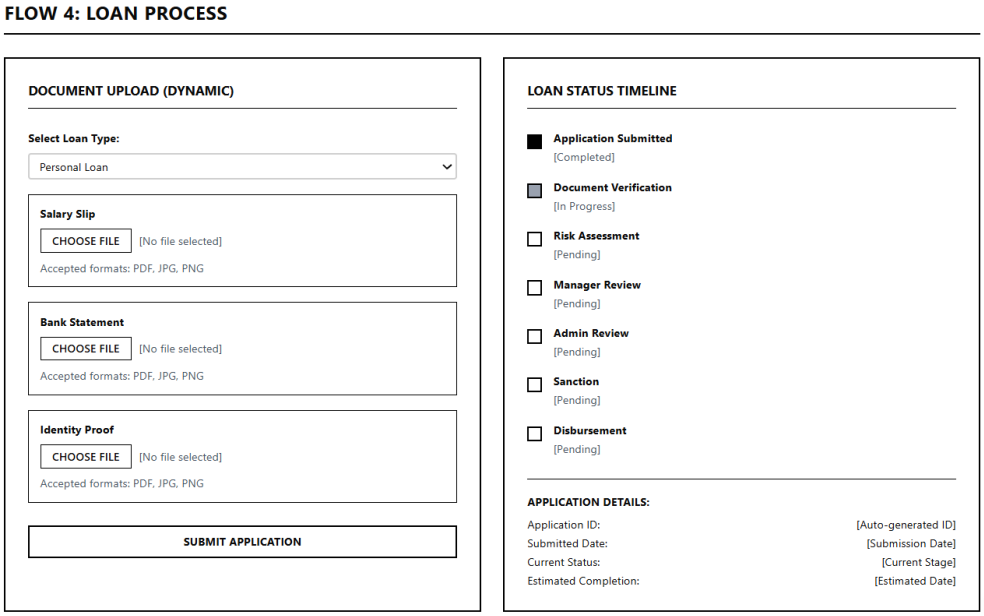
7.3.Customer's KYC Process



7.4.Available loans to browse



7.5.Loan Process



7.6.EMI and repayment

FLOW 7: EMI & REPAYMENT

EMI DASHBOARD

LOAN AMOUNT	EMI AMOUNT	REMAINING BALANCE	NEXT DUE DATE	
[Amount]	**[EMI]**	**[Balance]**	**[Date]**	PAY EMI

EMI PAYMENT

CURRENT EMI DETAILS:

Installment Number:	[Number]
Due Date:	[Date]
EMI Amount:	[Amount]
Penalty (if any):	[Penalty]
Total Payable:	**[Total]**

Payment Method:

[Select Payment Method]

Payment Amount:

[Amount]

PROCEED TO PAYMENT

LOAN SUMMARY

Principal Amount: **[Amount]**	Interest Rate: **[Rate %]**
Loan Tenure: **[Months]**	Total Interest: **[Amount]**
Total Payable: **[Amount]**	Amount Paid: **[Amount]**

EMI HISTORY

INSTALLMENT #	DUE DATE	EMI AMOUNT	PAID AMOUNT	PAYMENT DATE	PENALTY	STATUS
1	[Date]	[Amount]	[Amount]	[Date]	[Amount]	Paid
2	[Date]	[Amount]	[Amount]	[Date]	0	Paid
3	[Date]	[Amount]	-	-	-	Pending
4	[Date]	[Amount]	-	-	[Amount]	Overdue

SYSTEM NOTE:
Penalty applied for overdue EMIs as per policy

ESCALATION ALERT:
3 consecutive overdue EMIs trigger escalation

7.7. Manager's Dashboard

FLOW 5: MANAGER

MANAGER DASHBOARD

PENDING LOAN REQUESTS	PENDING SANCTION LETTERS
[Count]	**[Count]**

LOAN DECISION PANEL

LOAN DETAILS:

Application ID:	[ID]
Applicant Name:	[Name]
Loan Type:	[Type]
Loan Amount:	[Amount]
Risk Score:	[Score/100]

VERIFICATION SCORE:

Income Score:	[Score/25]
Employment Score:	[Score/25]
Credit Score:	[Score/25]
Document Score:	[Score/25]
Total Score:	**[Total/100]**

VIEW FULL APPLICATION

APPROVE (≤ 15 LAKHS)

FORWARD TO ADMIN (> 15 LAKHS)

REJECT

SANCTION LETTER REVIEW

LETTER DETAILS:

Letter ID:	[ID]
Applicant:	[Name]
Sanctioned Amount:	[Amount]
Interest Rate:	[Rate %]
Tenure:	[Months]

[SANCTION LETTER PREVIEW]

To,

[Applicant Name]
[Address]

Subject: Loan Sanction Letter - [Loan ID]

Dear [Name],

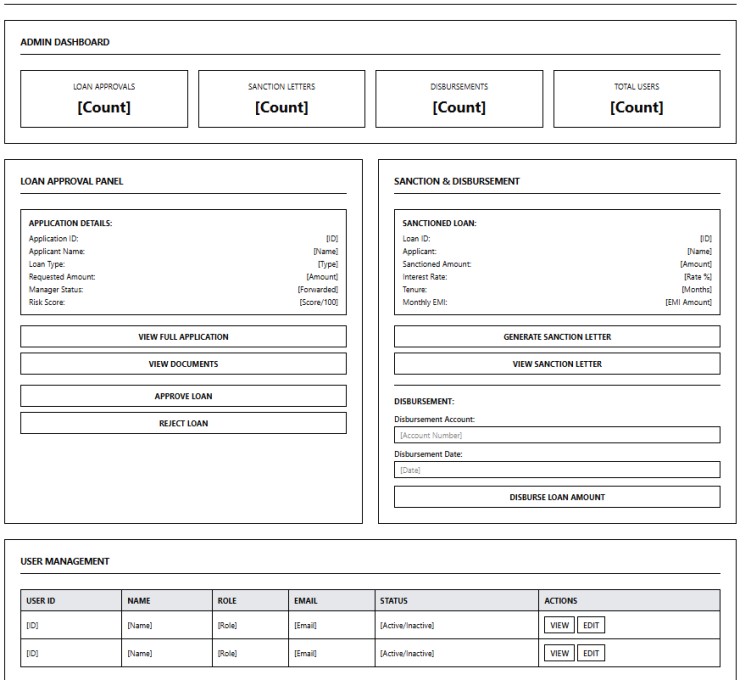
We are pleased to inform you that your loan application has been approved...

VIEW SIGNED LETTER

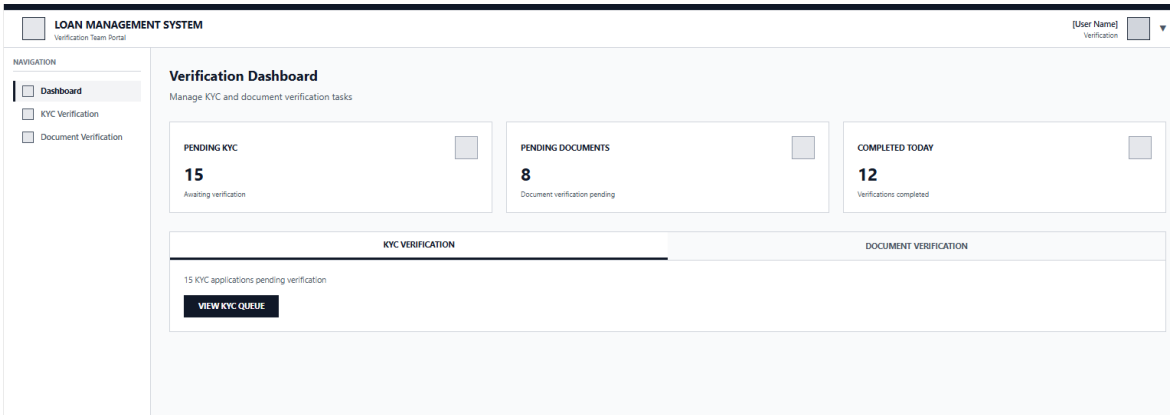
APPROVE

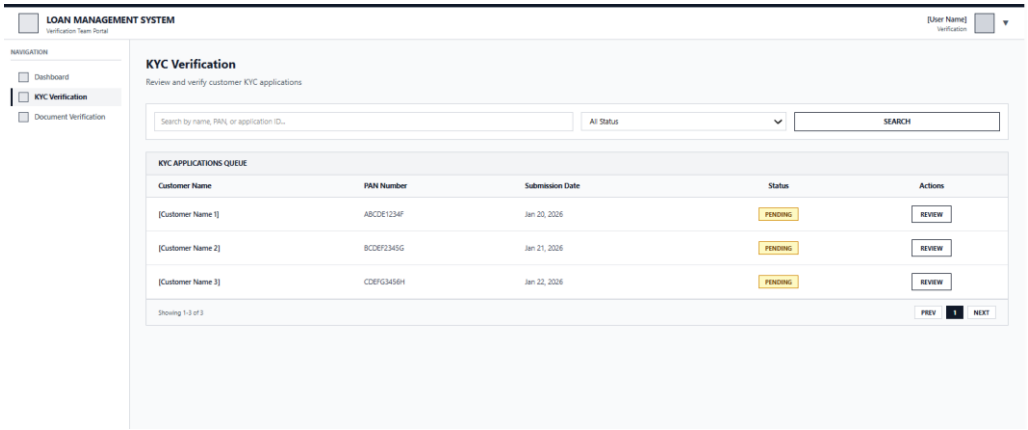
REJECT

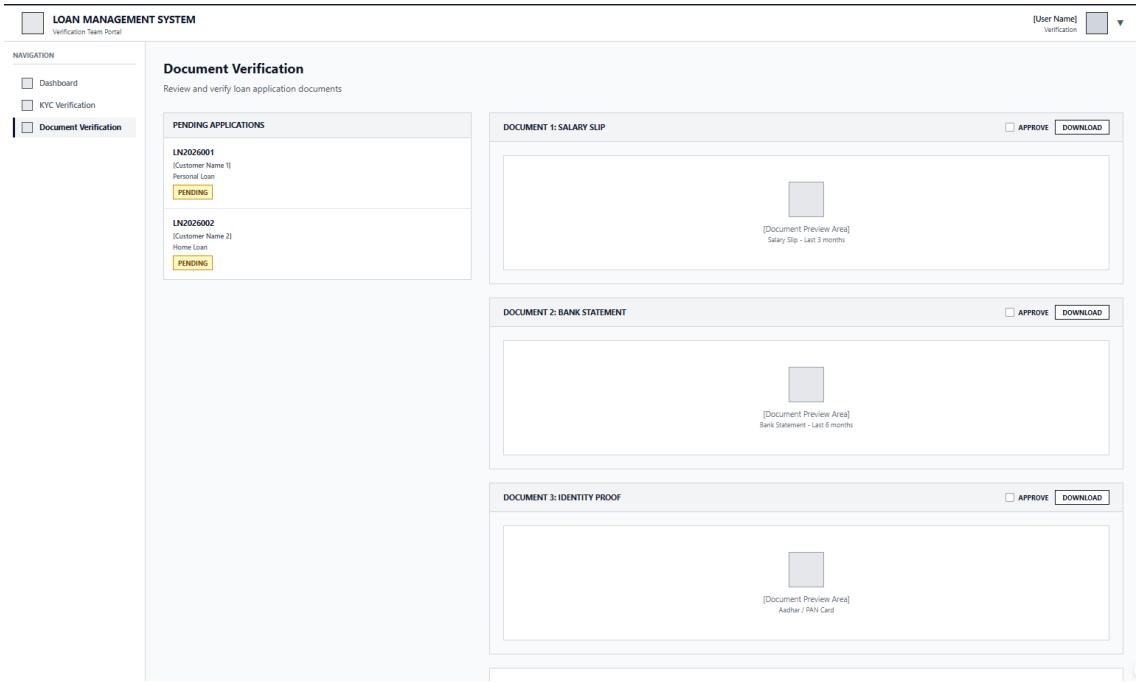
7.8. Admin's Dashboard



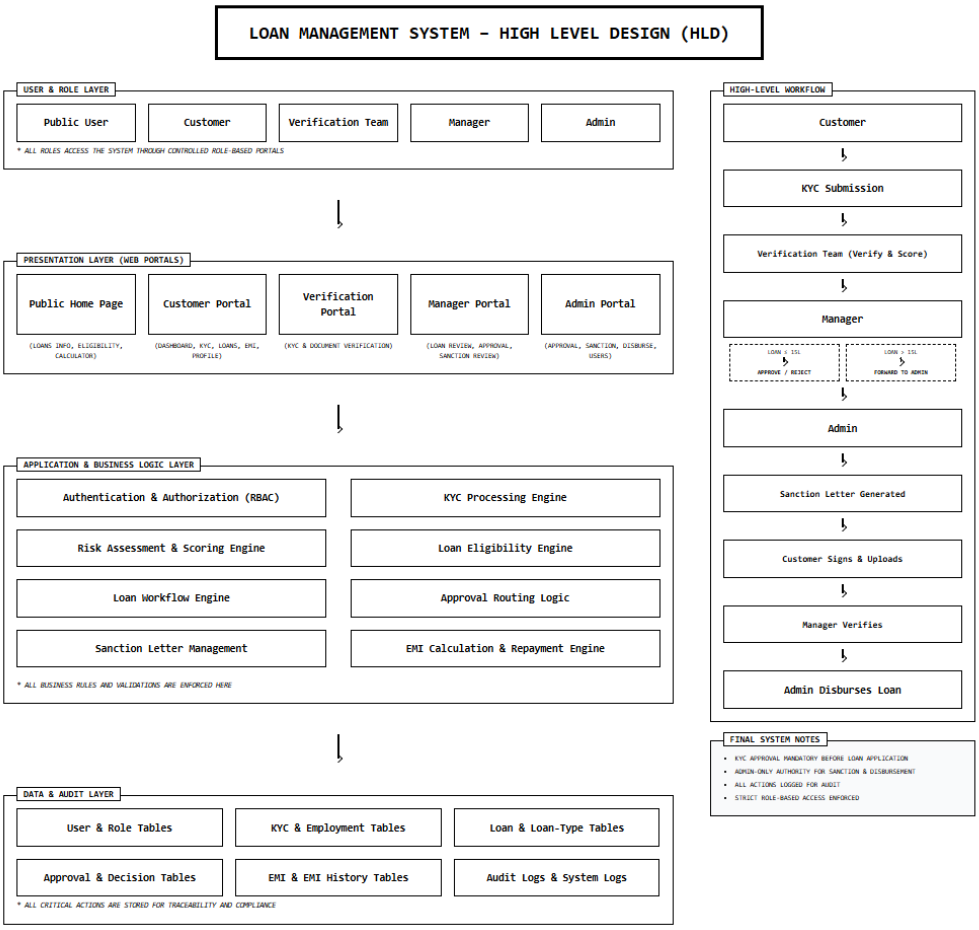
7.9 Verification Team Dashboard







7.10. Entire Flow of HLD



8. Conclusion

The Role-Based Loan Management System (LMS) designed in this project provides a comprehensive, secure, and enterprise-grade solution for managing the complete loan lifecycle. The system successfully digitizes customer onboarding, enforces mandatory KYC compliance, and ensures controlled loan processing through clearly defined roles and responsibilities.

By implementing strict role-based access control among Customer, Verification Team, Manager, and Admin, the LMS prevents unauthorized actions and maintains governance standards similar to real-world banking systems. The inclusion of an internal CIBIL-like risk assessment engine enables data-driven loan eligibility decisions, reducing manual bias and improving approval accuracy.

The system further ensures operational transparency through detailed loan status tracking, sanction letter management, and controlled disbursement. EMI scheduling, repayment tracking, penalty handling, and comprehensive EMI history provide full visibility into loan servicing, while audit and logging mechanisms ensure accountability.

8.1 Future Enhancements

Although the current LMS implementation fulfills all core functional and non-functional requirements, the system can be further enhanced to increase efficiency, intelligence, and user experience.

1. Integration with External Credit Bureaus

The internal CIBIL-like scoring engine can be enhanced by integrating with external credit bureaus to fetch real-time credit scores, providing more accurate risk assessment and decision-making.

2. Online EMI Payment Gateway

Integration with secure payment gateways (UPI, net banking, debit cards) would allow customers to pay EMIs directly through the platform, improving convenience and reducing manual intervention.

3. AI-Based Risk & Fraud Detection

Machine learning models can be introduced to analyze user behavior, transaction patterns, and repayment history to detect fraud and predict loan defaults more accurately.

4. Multi-Loan & Co-Applicant Support

Future versions can support multiple concurrent loans per customer and allow co-applicants or guarantors, aligning the system with advanced retail banking use cases.

5. Advanced Notifications & Alerts

Automated email and SMS notifications for KYC status, loan approvals, EMI due dates, overdue alerts, and escalation warnings can improve customer engagement and compliance.

6. Mobile Application Support

Developing native Android and iOS applications would improve accessibility and allow users to manage loans and EMI payments on the go.

7. Regulatory & Compliance Reporting

Automated generation of compliance and regulatory reports would help financial institutions meet statutory requirements efficiently.